Blessed G/Michael catholic school

First Biology final examination for grade 10th

Name Betanya Wondemagegnehu section 10th B roll number 12

Jan/2018 Time allowed 50 minutes

Instruction I Matching

Match column A with Column B

Column A	Column B
1. Contractile protein A C	A .Serum albumin
2. Respiratory pigment D	B .Enzymes
3. Chemical messengers BE	C .Actin
4. Transport protein B A	D .Myoglobin
5. Structural protein F	E .Hormones
	F .Elastin

Instruction:- II Choose the best answer among the given alternatives.

C 6, Short segment of DNA has 80 A thymine and 80 C guanine residues .The total number of nucleotides in this DNA fragment is ?

A .160 B .40 C .320 D .640

B 7. Which molecule carries amino acids to ribosomes during protein synthesis ?

A. m-RNA B .t -RNA C .r-RNA D .DNA

A 8. Which of the following is true about phosphodiester linkage ?

A. A 5 ' Phosphate group of one nucleotide unit is joined to the 3'hydroxyl group (sugar) of the next nucleotide

B. A 3 ' Phosphate group of one nucleotide unit is joined to the 5'hydroxyl group (sugar) of the next nucleotide

C. A 5 ' Phosphate group of one nucleotide unit is joined to the 5'hydroxyl group (sugar) of the next nucleotide

D .None

B 9. The two strands of a DNA double helix are joined by ?

A .Covalent bond B .Hydrogen bond C .Ionic bond D .Phosphodiester bond

A 10. Identify the purine base of nucleic acid in the following ?

A .Cytosine B .Thymine C .Uracil D .Adenine

A-U
T

B 11. RNA is the genetic material in ?

A .Some viruses and prokaryotes C .Some viruses only

B. Some prokaryotes only D .Some Eukaryotes

A 12. Lipids are important for the absorption and transport of fat soluble vitamins such as ?

A .Vitamin D B .Vitamin A C .Vitamin k D .All

A 13. Lipids play a role in the formation of myelin sheath which surrounds ?

A .Muscle fibers B .Nerve fibers C .Blood vessels D .None

C 14. Which of the following is not a macro nutrient ?

A .RNA B.DNA C. Salt D .Protein

D 15. Polymerization reaction which polysaccharides are synthesized from monosaccharaides ?

A .Require the formation of glycosidic bonds between monosaccharaides

B .Are hydrolytic reactions

C .Results in the formation of water

D .A & C

A 16. The fats and oils are respectively rich in ?

A .Saturated and unsaturated fatty acids C .Saturated and saturated fatty acids

B. Un Saturated and unsaturated fatty acids D .None

B D 17. Which one of the following is the major storage form of carbohydrate ?

A .Starch B .Glycogen C .Cellulose D . A & B

B 18. Which sugars are present in sucrose ? mal-2G FtG

A .Glucose and glucose C .Xylose and Glucose

B .Glucose and Fructose D .Glucose and Galactose - lactose

A 19. Which of these is a non reducing sugar ?

A .Glucose B .Sucrose C. Lactose D. Maltose

B 20. A carbohydrate found only milk is ?

A .Fructose B .Lactose C. Maltose D. Sucrose

B 21. Lipids is formed by the condensation reaction of ?

A. Fatty acid and amines C. Fatty acids and amino acids

B. Fatty acid and alcohols D. Carbon and hydrogen

D 22. Which of the following is a major function of lipids ?

A .Energy storage C. Structural component of cell membrane

B .Hormone synthesis D .All

C 23. Which of the following lipids is the most abundant lipid found in adipose tissues ?

A .Triglyceride B .Phospholipids C .Wax D . Sphingo lipids

most abundant PolySaccharide = cellulose

or [DNA degenerate after formation of protein & They carry code for many proteins

Omega 3 & 6 → Saturated

Good Cholesterol - LDL

Centromere - responsible for mitotic spindle

A 24. Which of the following lipids is commonly known as good cholesterol ?

A. LDL B. HDL C. Triglyceride D. Phospholipids

A 25. Omega 3 and omega 6 fatty acids are examples of ?

A .Poly unsaturated fatty acids C. Monounsaturated fatty acids

B .Saturated fatty acids D. All

D 26. The intake of excessive dietary lipids can contribute to the development of

A .Obesity B .Cardiovascular disease C .Type two diabetes D. All

A 27. All are characteristics of DNA except ?

A .They carry code for many proteins C .They are stable ✓

B .They degenerate after the formation of proteins D .They are double stranded ✓

D 28. Which of the following are present in the nucleotides ?

A .Sugar B .Phosphate C .Nitrogen containing bases D. All

A 29. The primary structure of proteins is maintained by ?

A .Peptide bond B .Hydrogen bond C .Ionic bond D .Disulfide bond

A 30. The peptide that contains two peptide bonds and three amino acids is ?

A .Dipeptide B .Tri peptide C .Tetra peptide D .Poly peptide

B 31. What makes phospholipids molecules arrange themselves into a bilayer in water medium ?

A .The length of fatty acid chains C .Their high solubility in water

B .The presence of hydrophilic and hydrophobic portions D .The double bonds in fatty acids

B 32. How many molecules of water can be produced from 200 moles of glucose joined together by a glycosidic bond to form poly saccharides ?

A.99 B.200 C .199 D .198

A 33. The most abundant poly saccharide in nature is ?

A .Cellulose B .Starch C. Glycogen D. Chitin

D 34. Which complex sugar is structural component of cell wall of fungi?

A .Cellulose B .Starch C. Glycogen D. Chitin

C 35. Where does DNA replication occur in the cell cycle ?

A .G1 phase B .G2 phase C .S phase D .M phase

B 36. What is the correct order of the main stages of mitosis ?

A .G1 →G 2→S →Mitosis →Cytokinesis C. G1 →G 2→S →Cytokinesis → Mitosis

B . G1 →S→G2 →Mitosis →Cytokinesis D . S →G 1→G2→Mitosis →Cytokinesis

C 37. What happens during anaphase of mitosis ?

A .Chromosomes condense and the nuclear envelope breaks down

B .Chromosomes align at the center

C .Sister chromatids separate and move to opposite poles

D .The cell divides into two daughter cells

A 38. Which cell structure is responsible for forming the mitotic spindle ?

A .Centromere B .Centrosome C .Kinetochore D .Cleavage furrow

A 39. The longest phase of interphase is ?

A .G1 phase B .G2 phase C .S phase D .M phase

D 40. What is the role of the spindle during mitosis ?

A .It makes the chromosomes visible

B .It breaks down the nuclear envelope

C .It duplicates the DNA

D .It helps separate the chromosome

D 41. What event occurs during interphase ?

A .The cell grows B .DNA replicates C .ready to mitosis D .All

B 42. The shortest phase of mitosis is ?

A .Metaphase B. Anaphase C .Prophase D . Telophase

B 43. Suppose a cell with 46 chromosomes undergo mitosis how many chromosomes each daughter cell contains ?

A .23 B .46 C .92 30 D .48

Answer Sheet

Name Bitaniya Wondimagegnehu Section 10th B No 12

1.	A C	11.	B C	21.	B	31.	B	41.	D
2.	D	12.	A D	22.	D	32.	B C	42.	B
3.	E	13.	A B	23.	C A	33.	A	43.	B
4.	B A	14.	C	24.	A B	34.	D		
5.	F	15.	D	25.	A	35.	C		
6.	C	16.	A	26.	D	36.	B		
7.	B	17.	A D	27.	A B	37.	C		
8.	A	18.	B	28.	D	38.	A B		
9.	B	19.	A B	29.	A	39.	A		
10.	A D	20.	B	30.	A B	40.	D		